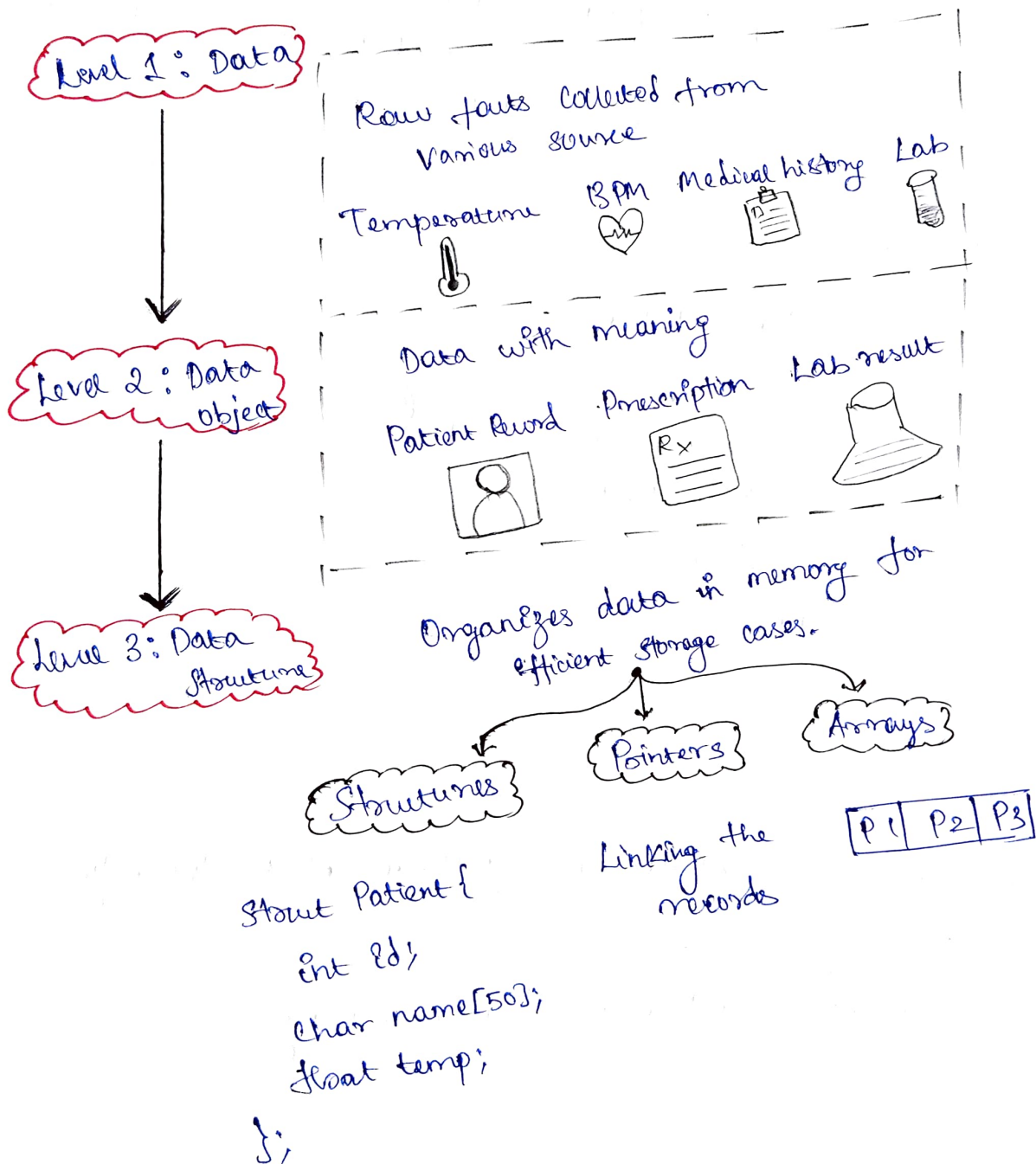
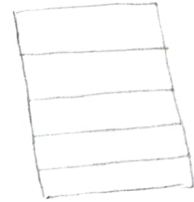


Question 1:- Smart healthcare system - Data organization



Level 4: ADT

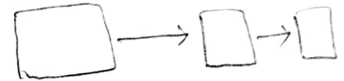
Stack: Used to make Undo operations in patient history



Queue: Patient Queue at OPD or LAB for processing



List: Maintain medical records / Prescription as a list



Level 5: Implementation

struct Patient{

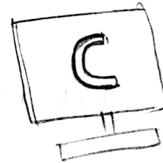
int id;

char name[50];

float temp;

float bp;

};



Outcome:

- Fast access
- Better memory Usage
- Data Integrity
- Scalability
- Improved decision making

Data Structures

Primitive DS

Ex: int, char, float
double & bool

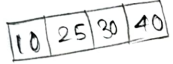
Traffic ID
Speed
Time
Signal Info

Non-primitive DS

Linear

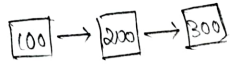
Non Linear

Arrays



Store traffic
Count per hour

Linked Lists

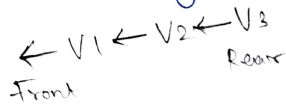


Maintain dynamic
List of vehicles

Queues

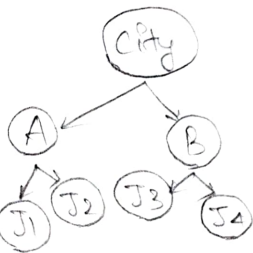
FIFO

Vehicle waiting
at traffic signal
or toll gate



Trees

Hierarchical Structure



Represent traffic zones &
Sub zones hierarchy

Graphs

Network of nodes

Road network modeling
Shortest path for routing
Traffic flow optimization

